Internet of Things

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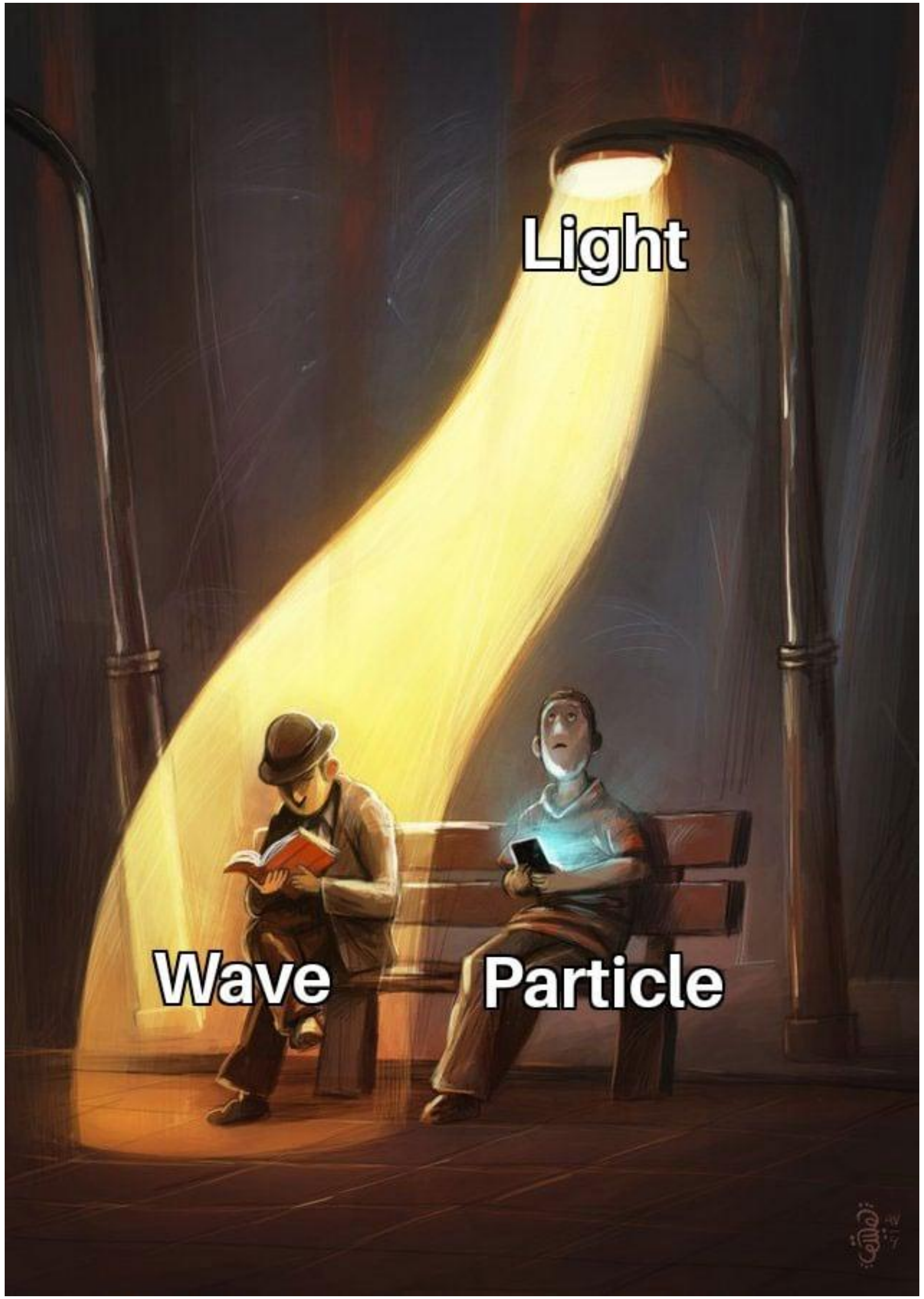
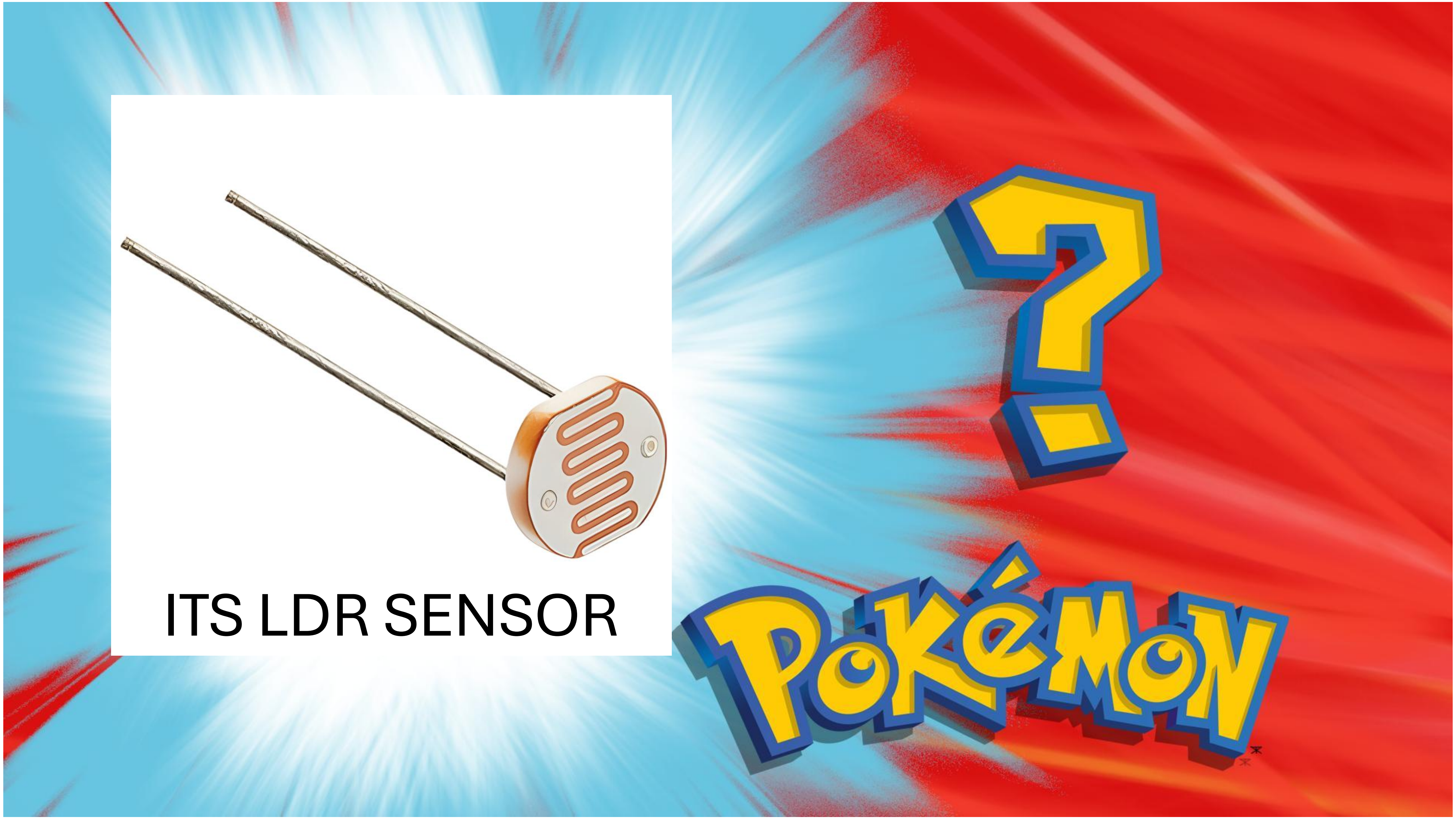
Problem

I want to find out how bright is this room? What sensor should I choose?


Light
Wave
Particle


POKéMON

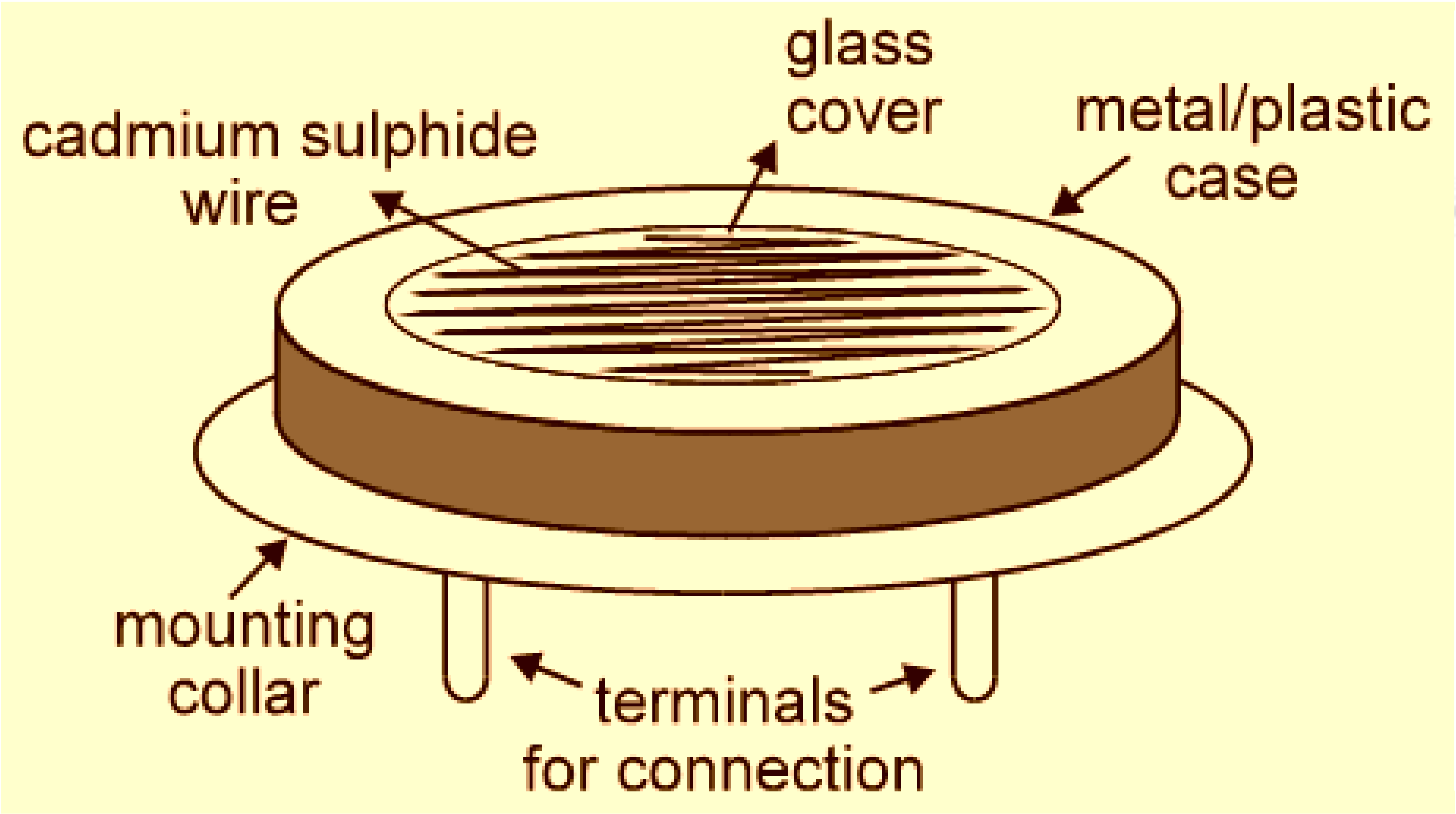

ITS LDR SENSOR
Pokémon

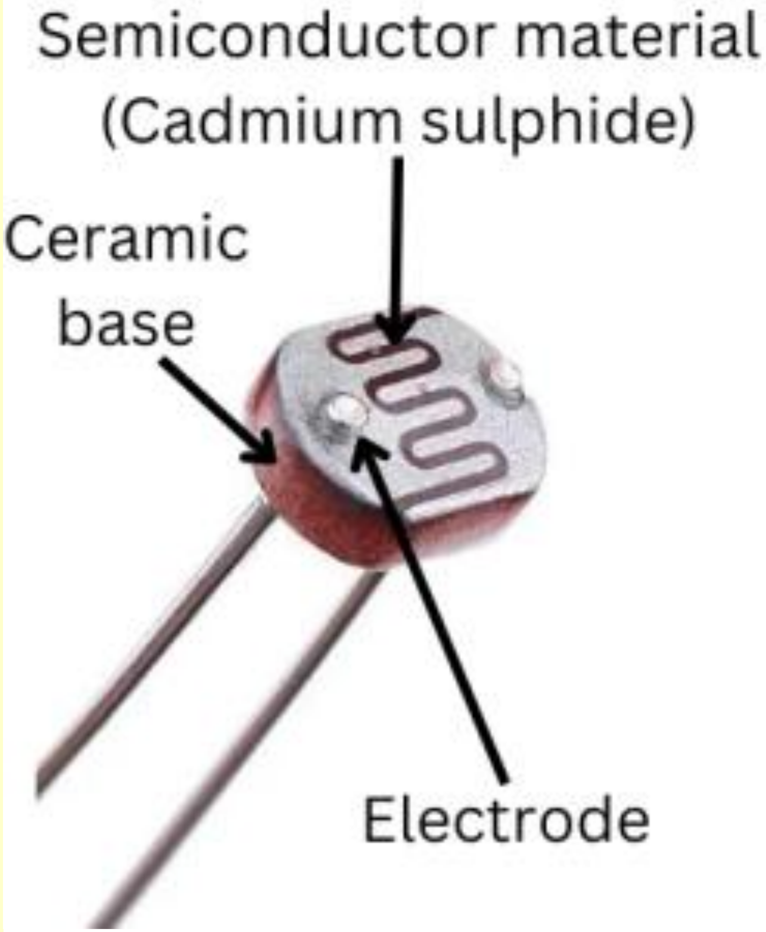
What is an LDR Sensor?

An LDR sensor, also known as a **Light Dependent Resistor**, is a passive electronic component that detects light in its environment.

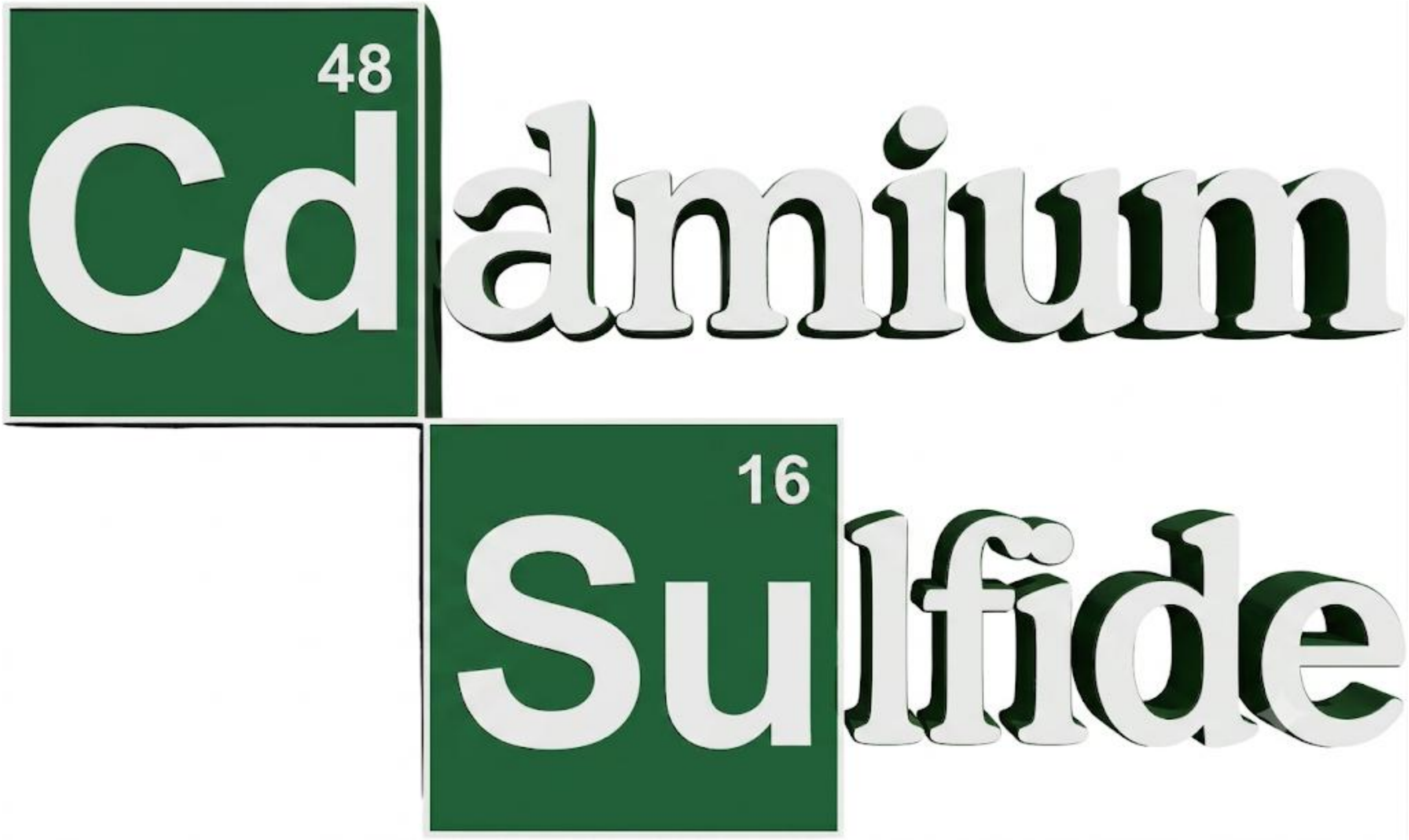
This remarkable device plays a crucial role in various applications by changing its electrical resistance based on the intensity of light it's exposed to.

The LDR sensor working principle relies on photoconductivity, which allows it to convert changes in illumination into electrical signals that can be measured and interpreted.





https://www.eeeguide.com/light-dependent-resistors-ldrs/


48
Cd
admium
16
Su
lfide

Components of an LDR Sensor

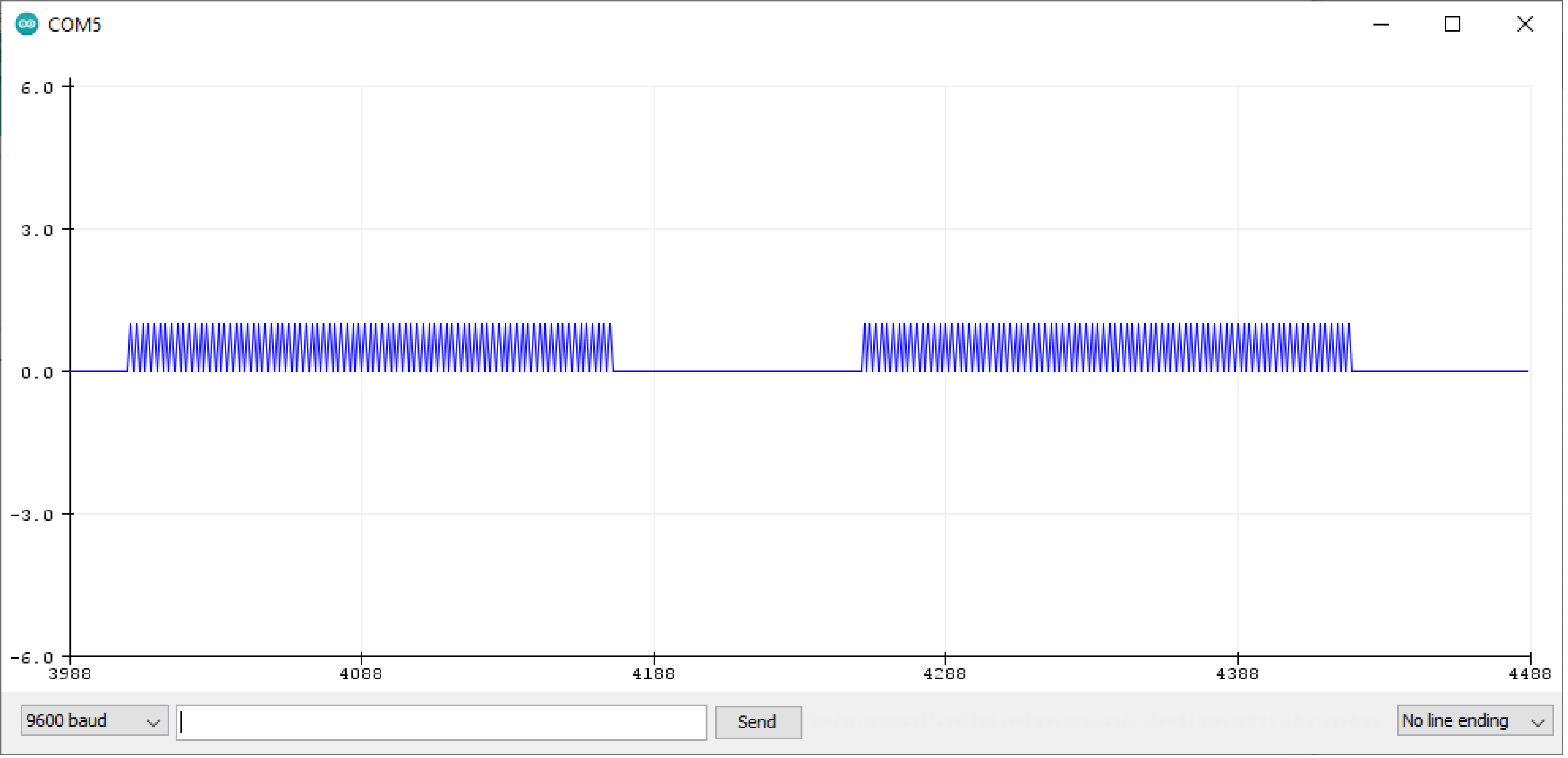
The structure of an LDR sensor is relatively simple yet effective. It consists of a light-sensitive semiconductor material, typically cadmium sulfide (CdS), deposited on an insulating substrate like ceramic.

The photosensitive material is arranged in a zig-zag pattern to maximize its surface area and sensitivity to light. Two metal contacts are placed at both ends of this pattern to create an electrical connection. A transparent coating is applied on top to protect the photosensitive material while allowing light to pass through.

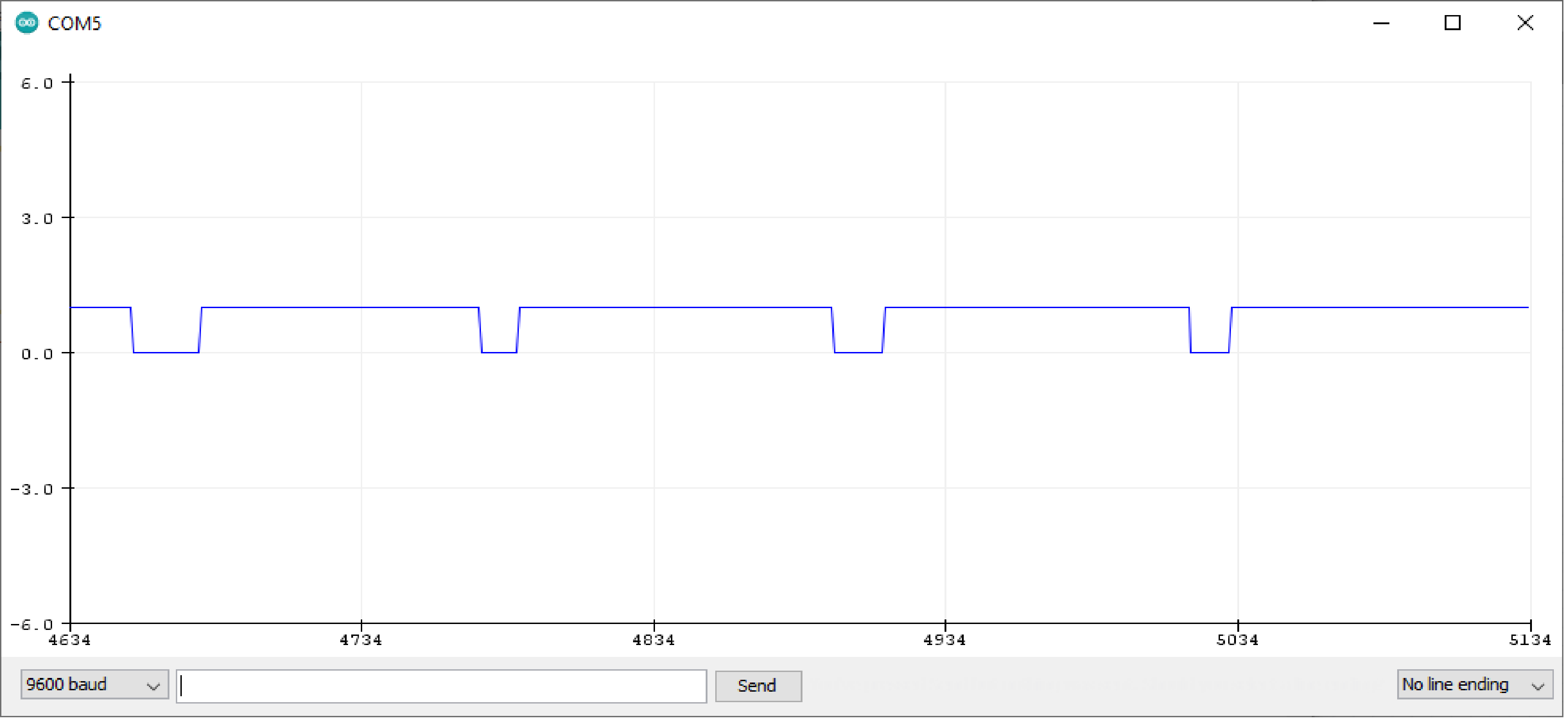
pinMode

- **OUTPUT:** this is to write data to an actuator, for example an LED.
- **INPUT:** in this case you're going to read data from the sensor. The value you'll get will be HIGH or LOW (binary).
- **INPUT_PULLUP:** This option is the same as INPUT (you read data from the sensor), but in addition to that, an internal pull up resistor – between 20k and 50k Ohm – is enabled, to keep the signal HIGH by default.

https://roboticsbackend.com/arduino-input_pullup-pinmode/




```
#define BUTTON_PIN 4
void setup() {
Serial.begin(9600);
pinMode(BUTTON_PIN, INPUT);
}
void loop() {
Serial.println(digitalRead(BUTTON_PIN));
delay(10);}
```

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```

Assignment

1. WAP to interface with the LDR Sensor.
2. WAP to interface LDR with turning the LED light ON and OFF based on the intensity of light.
3. WAP to dim or increase the LED light based on the values from LDR. (The Values recorded by LDR should be converted into **Byte.** Since the LED Values maximum in 255)

- Hint its an Analog Sensor.

Thanks